



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT
BS Computer Science
COURSE INFORMATION SHEET

Session:	Fall-2023
Course Title:	Computer Organization & Assembly Language
Course Code:	CS-215T
Credit Hours:	3
Semester:	3 rd
Pre-Requisite:	Digital Logic Design (CS-128T)
Instructor Name:	Mr. Shakir Karim Buksh
Email and Contact Information:	skbux@ssuet.edu.pk ; HS-16 (cubical-06)
WhatsApp Group:	COAL_Fall2023
Office Hours:	Tuesdays & Wednesdays (10:00am to 11:30am)
Mode of Teaching:	Synchronous/Asynchronous/Hybrid/Blended

COURSE OBJECTIVE:

The objective of this course is to explain the concepts of computer organization in relation with Intel 8086/88 instruction set architecture. The concepts of Assembly language programming for the Intel 8086/88 related to assembling and linking programs, addressing modes, program control flow, arithmetic and logical operations, ASCII and BCD arithmetic, procedure calls, and string manipulation will be discussed. Students will understand the design issues that can affect the performance of a computer system, including the elements and design concept of cache memory using different mapping techniques.

COURSE OUTLINE:

Fundamentals of Computer Design, Computer structure and function, Brief history of Intel x86 based Computer system, Processor Model (Registers and Program Segments), Computer Arithmetic Logic Unit, System Commands Program (Process) Control commands, Status Flags, Physical and Logical Address, Stack, Queue, and Basic Terminology, Introduction to Assembly programming, 8086/88 Programming model, Data Directives, Data Types and Definitions, Instructions and Operands Types, Addressing Modes, Data Movement Instructions, Control Transfer Instructions (JUMP and Loop instructions) and their types, Procedures using CALL and RET instructions, Concept of Interrupt and Polling, Machine – level representation, Arithmetic and Logical instructions, Floating Point Arithmetic, Booth Algorithm, BCD and ASCII Arithmetic provision in x86 assembly language, String Manipulation Instructions, Computer Memory and its Types, Memory Hierarchy, Memory Decoding Techniques, Cache Memory, Cache Memory Mapping, Direct Mapping Techniques, Associative Mapping Techniques, Set-Associative Technique, Pipelining concept, Pipelining Hazards, Control and Data Hazards regarding Pipelining, Branch Prediction as remedy for control hazards, Static and Dynamic algorithms of branch prediction, I/O and Storage Systems, I/O Architecture, RAID and its Level, Advanced / Recent Topics.



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COURSE LEARNING OUTCOMES (CLOs):

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
1	Explain the concepts of computer organization in relation with Intel 8086/88 instruction set architecture.	PLO_1 (Academic Education)	C2 (Understanding)
2	Describe the design issues affecting the performance of a computer system.	PLO_2 (Knowledge for Computing Problem)	C2 (Understanding)
3	Apply the elements and design concepts of cache memory using different mapping techniques to improve the performance of computer system.	PLO_3 (Problem Analysis)	C3 (Applying)

RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO1	CLO2	CLO3
Quizzes	3 (7%)	3 (12%)	4 (12%)
Assignments	3 (7%)	3 (12%)	4 (12%)
Mid	30 (73%)	-----	-----
Final	5 (12%)	20 (77%)	25(76%)
Total	41 (41%)	26 (26%)	33 (33%)

GRADING POLICY:

Assessment Tools	Percentage	Marks
Quizzes	10%	10
Assignments	10%	10
Midterm Exam	30%	30
Final Exam	50%	50
TOTAL	100%	100

Recommended Book:

- Stallings, William, *Computer Organization & Architecture: Design for Performance*, 11th edition, Pearson Education / Prentice Hall, 2019.
- Muhammad Ali Mazidi, and Janice Gillispie Mazidi, *The 80x86 IBM PC and Compatible Computers (Vol. I & II) Assembly Language, Design, and Interfacing*, 4th edition, Pearson Education, Inc. Upper Saddle River, New Jersey 07458

Reference Books:

- Null, Linda, *The Essentials of Computer Organization & Architecture*, 5th edition, Jones & Bartlett Learning, 2018.

Benchmarking:

- [Capital University of Science and Technology, Pakistan](#)
- [Arizona State University, USA](#)



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LECTURE PLAN

Course Title: Computer Organization & Assembly Language

Course Code: CS-215T

Week #	Week Dates	Topics	Required Reading	Key Date
1	09-10-2023 To 13-10-2023	Fundamentals of Computer Design DLD Revisit, Number System with emphasis on hexadecimal system, Computer structure and function, Brief history of Intel x86 based Computer system, Processor Model (Registers and Program Segments).	Material uploaded on course website **WS-Chap 1: Pg. 2 – 10	
2	16-10-2023 To 20-10-2023	Computer Arithmetic Logic Unit System Commands, Program (Process) Control commands, Status Flags, Physical and Logical Address, Stack, Queue, and Basic Terminology	* LN-Chap 1: Pg. 12 – 24 ***MZ-Chap 1: Pg. 23 – 40	
3	23-10-2023 To 27-10-2023	Introduction to Assembly Language 8086/88 Programming model, Data Directives, Data Types and Definitions, Instructions and Operands Types	***MZ-Chap 2: Pg. 50 – 62 ***MZ-Chap 2: Pg. 68 – 75	Practice Problems
4	30-10-2023 To 03-11-2023	Addressing Modes Addressing requirement, Types of Addressing Modes and their uses Data Movement Instructions	***MZ-Chap 1: Pg. 41 – 45	Assignment #1
5	06-11-2023 To 10-11-2023	Machine-Level Representation Machine instructions using Intel x86 instruction set Control Transfer Instructions JMP instructions and their types, Implementation of Loop in Assembly Language.	Handout ***MZ-Chap 2: Pg. 63 – 67	Practice Problems
6	13-11-2023 To 17-11-2023	Control Transfer Instructions (cont...) Procedures using CALL and RET instructions Concept of Interrupt and Polling, Arithmetic instructions Add, Sub, Mul, and Div.	***MZ-Chap 2: Pg. 66 – 68 **WS-Chap 10: Pg.340 ***MZ-Chap 3: Pg. 83 – 92	Quiz #1
7	20-11-2023 To 24-11-2023	Booth Algorithm for multiplication, Floating Point Arithmetic Logic Instructions Shift and Rotate instructions,	**WS-Chap 10: Pg.350 ***MZ-Chap 3: Pg. 93 – 103	Practice Problems
8	27-11-2023 To 01-12-2023	Character Codes BCD & ASCII with their operands and instructions (DAA, DAS, AAA, AAS, AAM, AAD)	***MZ-Chap 3: Pg. 104 – 113	



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9	Midterm Examination (04-12-2023 to 09-12-2023)			
10	11-12-2023 To 15-12-2023	String Manipulation Instructions String movement, String comparison, String Scanning instructions Computer Performance Measures Amdahl's Law, Little's Law Computer Function, Hardware and Software approach	***MZ-Chap 6: Pg. 184 – 189 **WS-Chap 2: Pg. 56 **WS-Chap 3: Pg. 81 – 89	Assignment #2
11	18-12-2023 To 22-12-2023	Computer Memory Memory Hierarchy, Concept of Locality of reference, Computer Memory & its Types, Cache Memory, Level of Cache	**WS-Chap 4: Pg. 121 **WS-Chap 4: Pg. 128	
12	25-12-2023 To 29-12-2023	Cache Memory Mapping Direct Mapping Techniques, Associative Mapping Techniques, Set-Associative Mapping Technique, Replacement Algorithms, Cache coherence	**WS-Chap 4: Pg. 134	Quiz #2 Assignment #3
13	01-01-2024 To 05-01-2024	Pipelining Concept of Pipelining, Pipelining Hazards, Control, Resource, and Data Hazards	**WS-Chap 14: Pg. 507 – 509	
14	08-01-2024 To 12-01-2024	Remedy for Control Hazards Branch Prediction technique, Static and Dynamic algorithms of branch prediction	Reference material available on course website	Quiz #3
15	15-01-2024 To 19-01-2024	Concept of RISC and CISC RISC vs CISC Redundant Array of Independent Disks (RAID) Uses and Types of RAID	*LN-Chap 7: Pg. 302 – 308	
16	22-01-2024 To 26-01-2024	Advanced Topics Parallel and Multiprocessor Organization, Introduction to cloud computing and GPU	*WS-Chap 17: Pg. 615 – 617 *WS-Chap 19: Pg. 691 – 700	
Final Examination (29-01-2024 to 10-02-2024)				

*LN: Linda Null; **WS: William Stallings; ***MZ: Muhammad Ali Mazidi